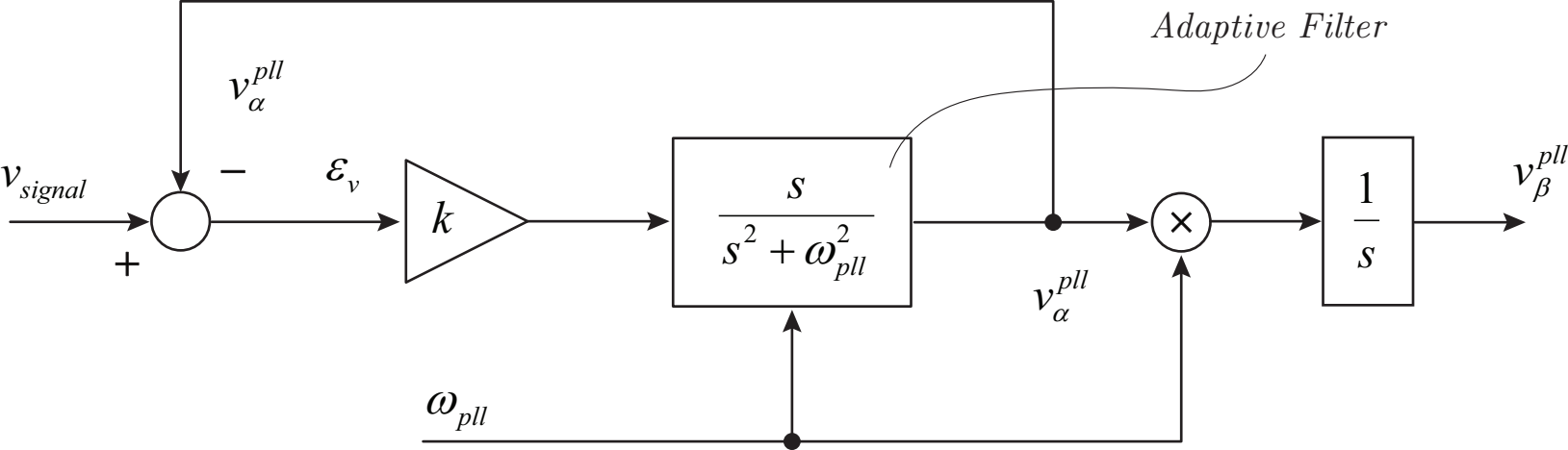
Adaptive Filter
v_{signal}
v_α^{pll}
$-$
$+$
ε_v
k
$\frac{s}{s^2+\omega_{pll}^2}$
v_α^{pll}
$\times$
$\frac{1}{s}$
v_β^{pll}
ω_{pll}